Stores `r` in the asynchronous result associated with `*this` if `ResultType` isn't `void`.

Postconditions

The asynchronous result associated with `*this` is *ready* with a stored value. Any threads blocked waiting for the asynchronous result are unblocked.

Throws

An exception of type `std::future_error` with an error code of `std::future_errc::promise_already_satisfied` if the asynchronous result already has a stored value or exception. Any exceptions thrown by the copy-constructor or move-constructor of `r`.

Synchronization

Multiple concurrent calls to `set_value()`, `set_value_at_thread_exit()`, `set_exception()`, and `set_exception_at_thread_exit()` are serialized. A successful call to `set_value()` happens-before a call to `std::future<ResultType>::get()` or `std::shared_future<ResultType>::get()`, which retrieves the value stored.

Std::Promise::Set_Value_At_Thread_Exit Member Function

Stores a value in the asynchronous result associated with `*this` without making that result *ready* until the current thread exits.

Declaration

```
void promise<void>::set_value_at_thread_exit();
void promise<R&>::set_value_at_thread_exit(R& r);
void promise<R>::set_value_at_thread_exit(R const& r);
void promise<R>::set_value_at_thread_exit(R&& r);
```